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SUPPLEMENTARY MATERIAL

The SST in Its Quasi-Static Limit

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Abstract
The scalar-tensor transition (SST) is the four-term covariant action of Gravity Field Dynamics, with kinetic function $F(y) = y - 2\sqrt{y} + 2\ln(1+\sqrt{y})$. We derive, exactly, what it gives in the quasi-static weak-field limit in which galaxies are observed, reading the action as the framework does: Φ is the Newtonian potential of the metric, the curvature term carries the tensor sector, and Φ's response is carried by F. Varying the action gives the field equation $\nabla\cdot[\mu(|\nabla\Phi|/a_0)\nabla\Phi] = 4\pi G\rho$ with $\mu(x) = x/(x+1)$, equivalently the capacity form with $f(x) = (21x+13)/(x+1)$. In spherical symmetry it integrates exactly to the conserved flux $r^2\mu g_t = GM$ and the spherical field equation $g_s = g_t^2/(a_0+g_t)$, whose forward solver and mass decoder are exact inverses. We derive the Newtonian and deep limits, the baryonic Tully–Fisher relation $v^4 = GMa_0$, the bound $a_0\mu < a_0$ on the TetB share, the location of the transition ring, the phantom density, and the well-posedness of the field equation. With the single potential in both metric components, $\gamma = 1$: a point mass deflects light by $4GM/(bc^2)$, the deep field by $2\pi\sqrt{GMa_0}/c^2$, and lensing mass equals dynamical mass.

Keywords: scalar-tensor transition, covariant action, AQUAL, constitutive law, radial acceleration relation, Tully–Fisher, lensing, proof

Setting

The SST is the action [1, 2, 3]

$$S = \int d^4x\,\sqrt{-g}\,\Big[\frac{R}{16\pi G} - \frac{a_0^2}{8\pi G}\,F(y) - \tfrac{1}{4}e^{-2\Phi/c^2}F_{\mu\nu}F^{\mu\nu} + \mathcal{L}_{\text{matter}}\Big], \tag{1}$$

with $y = |\nabla\Phi|^2/a_0^2$ and $F(y) = y - 2\sqrt{y} + 2\ln(1+\sqrt{y})$. F is the unique energy of the constitutive law $\mu(x) = x/(x+1)$, $F'(y) = \mu(\sqrt{y})$, $F(0) = 0$ [4]; μ is derived from the exchange symmetry of the two roots of the tetrad equation [5, 6]. We use the framework's reading of the action, as stated in the archive:

(i) Φ is the Newtonian potential of the metric, "not a new auxiliary field; the partition modifies its response, not its identity" [1], with the torsion share $1-\mu$ carried inside F [2].

(ii) The curvature term carries the tensor sector, so gravitational waves travel at c, and Φ's response is carried by F, the scalar sector "decoupling from the metric dynamics" [2]. (The Remark after Theorem 6 shows why this is needed.)

(iii) In the static weak field, matter enters through the source term $-\rho\Phi$, the weak-field form of minimal coupling when Φ is the metric potential, and $e^{-2\Phi/c^2} \to 1$ [7].

(iv) μ applies to the net field of a galaxy, not to each mass; a bound system much smaller than its host, such as the solar system, is pure TetA, that is, general relativity [8, 9].

Naming. "SST" names the action; the spherical, static relation $g_s = g_t^2/(a_0+g_t)$ is its spherical field equation [8]. The documentation calls the capacity function $f(x) = (21x+13)/(x+1)$ "the field equation" [10]; Theorem 2 shows that the two are one equation.

The Field Equation

Theorem 1. *In the static weak field the SST reduces to* $\int d^3x\,[-(a_0^2/8\pi G)F(|\nabla\Phi|^2/a_0^2) - \rho\Phi]$, *and varying* Φ *gives*

$$\nabla\cdot\Big[\mu\Big(\frac{|\nabla\Phi|}{a_0}\Big)\nabla\Phi\Big] = 4\pi G\rho. \tag{2}$$

Proof.
$\partial F/\partial(\nabla\Phi) = F'(y)\,2\nabla\Phi/a_0^2 = 2\mu\nabla\Phi/a_0^2$, and $(a_0^2/8\pi G)(2/a_0^2) = 1/(4\pi G)$. The Euler–Lagrange equation is $(4\pi G)^{-1}\nabla\cdot(\mu\nabla\Phi) - \rho = 0$, the Bekenstein–Milgrom form [14]. □

Theorem 2 (the capacity form). *With* $\sigma(x) = (x-1)/(x+1)$, *the capacity function* $f(x) = 17 + 4\sigma(x) = (21x+13)/(x+1)$ *takes the values* $13, 17, 21$ *at* $x = 0, 1, \infty$, *and* $\mu = (f-13)/8$. *So Eq. (2) is* $\nabla\cdot[\tfrac{1}{8}(f-13)\nabla\Phi] = 4\pi G\rho$, *the form of Paper VIII and the documentation.*

Proof. $17(x+1) + 4(x-1) = 21x + 13$, and $(21x + 13 - 13x - 13)/8 = x$. □

The Spherical Field Equation

Theorem 3. *In spherical symmetry, with $g_t = |\Phi'(r)|$ and $g_s = GM(<r)/r^2$, Eq. (2) integrates exactly to the conserved flux $r^2\mu(g_t/a_0)\, g_t = GM(<r)$, that is,*

$$g_s = \frac{g_t^2}{a_0 + g_t}, \qquad g_t = \tfrac{1}{2}\Big[g_s + \sqrt{g_s^2 + 4g_s a_0}\Big], \tag{3}$$

the second being the unique positive root. With $g_t = v^2/r$, the mass decoder

$$M_{\rm dec}(r) = \frac{rv^4}{G(a_0 r + v^2)} \tag{4}$$

is the exact inverse of the forward solver.

Proof. For $\Phi = \Phi(r)$, $\nabla\cdot(\mu\nabla\Phi) = r^{-2}(r^2\mu\Phi')'$, and $4\pi r^2\rho = M'(r)$, so the divergence theorem gives the flux with no curl term. Substituting $\mu = x/(x+1)$, $x = g_t/a_0$, gives the quadratic $g_t^2 - g_s g_t - g_s a_0 = 0$, whose other root is negative. The decoder is $r^2 g_s/G$ at $g_t = v^2/r$. □

This is the relation fitted to the SPARC rotation curves with no parameter in μ [7], solved forward in Paper X [11] and inverted in the decoding-mass letter [12].

Limits and Scales

Theorem 4. *(i) As $g_s \to \infty$, $g_t = g_s + a_0 - a_0^2/g_s + O(g_s^{-2})$: Newtonian. (ii) As $g_s \to 0$, $g_t = \sqrt{g_s a_0} + g_s/2 + O(g_s^{3/2})$. (iii) For a bounded mass M, $v^4 \to GMa_0$ exactly as $r \to \infty$: the baryonic Tully–Fisher relation. (iv) The TetB share of the field is $(1-\mu)g_t = a_0\mu(g_t/a_0) < a_0$. (v) The ring $x = 1$, where $\mu = \frac{1}{2}$, lies where $g_s = a_0/2$, at $r_1 = \sqrt{2GM/a_0}$; at $r = \sqrt{GM/a_0}$, where $g_s = a_0$, the field is at $x = \varphi = (1+\sqrt{5})/2$ with $\mu = 1/\varphi$. (vi) Outside a point mass the phantom density $\rho_{\rm ph} = (4\pi G r^2)^{-1}\, d(r^2 g_t)/dr$ has leading term $\sqrt{GMa_0}/(4\pi G r^2)$.*

Proof. (i), (ii): expansions of the forward solver. (iii): $v^4 = g_t^2 r^2$ with $g_s = GM/r^2$, and the limit of the solver. (iv): $g_t/(1+x) = a_0 x/(1+x)$. (v): $x = 1$ gives $g_s = 1/(1+1)\cdot a_0$; $g_s = a_0$ gives $x^2 = x + 1$. (vi): $r^2 g_s = GM$ is constant outside the mass, so the residual of the Newtonian flux is $r^2 g_t$, which grows as $\sqrt{GMa_0}\, r$. □

Theorem 5 (well-posedness). *The energy density $F(|p|^2/a_0^2)$ is strictly convex in $p = \nabla\Phi$ for $p \neq 0$, with Hessian eigenvalues 2μ across and $2(x\mu)'$ along the field, whose ratio $(x\mu)'/\mu = 1 + (1-\mu)$ lies in $[1, 2]$. So Eq. (2) is elliptic away from $\nabla\Phi = 0$, and its solution is unique for given boundary data.*

Proof. The energy-functional proof [4]; the eigenvalues are $2x/(x+1)$ and $2x(x+2)/(x+1)^2$. □

The Single Potential

Theorem 6. *Under reading (i), the weak-field metric carries the one potential in both components, $g_{00} = -(1 + 2\Phi/c^2)$, $g_{ij} = (1 - 2\Phi/c^2)\delta_{ij}$, so $\gamma = 1$, and the deflection of light at impact parameter b is $\theta = (2/c^2)\int g_\perp\, dz$. A Newtonian point mass gives $4GM/(bc^2)$; the deep field $g_t = \sqrt{GMa_0}/r$ gives the constant $2\pi\sqrt{GMa_0}/c^2$; and in general the lensing mass equals the dynamical mass $M_{\rm dyn}(r) = g_t r^2/G$. In a galaxy the factor $e^{-2\Phi/c^2}$ differs from 1 by about $2|\Phi|/c^2 \sim 10^{-6}$, so the fourth term does not enter the static dynamics.*

Proof. The deflection is $((1+\gamma)/c^2)\int \nabla_\perp\Phi\, dz$ with $\nabla_\perp\Phi = g_t\, b/r$; the integrals $\int b\, r^{-3} dz = 2/b$ and $\int b\, r^{-2} dz = \pi$ give the two values. A general-relativistic lens with mass profile $M_{\rm dyn}$ has $g = GM_{\rm dyn}/r^2 = g_t$, the same integrand. □

This is the content of Letter 3's statement that $\gamma = 1$ "because the single potential Φ governs both metric components", and of Paper I's single-standard postulate [13]. The deep-field deflection is Milgrom's [15].

Remark (the reading in (ii)). The static weak-field part of $R/16\pi G$ is $(8\pi G)^{-1}[(\nabla\Psi)^2 - 2\nabla\Phi\cdot\nabla\Psi]$, which on $\Psi = \Phi$ is the Newtonian energy $-(\nabla\Phi)^2/8\pi G$. If that part were counted for Φ in addition to F, the equations would give $\Psi = \Phi$ and $\nabla\cdot[(1+\mu)\nabla\Phi] = 4\pi G\rho$. The SST's field equation, Eq. (2), is the one in which Φ's response is carried by F alone, as the archive states; this fixes the reading, and Theorems 1–6 use it.

Scope

Theorems 1–5 are exact consequences of the SST in the quasi-static weak field, and Theorems 3–4 are what the SPARC rotation curves, the Tully–Fisher relation and the mass decoder test. Theorem 6 holds under the single-potential reading of Letter 3. The archive records what remains open [8]: the metric field equations, including the stress–energy of the F sector, have not been solved beyond this limit; $\mathcal{L}_{\rm matter}$ is specified here only in its weak-field form; the covariant meaning of y for time-dependent and strong fields is open; the documents describe the fourth term differently (disformal and needed for lensing; conformal with a cosmological role; an auxiliary vector), and the dependence of the fine-structure constant on the potential that it

implies is not discussed; and the match from the locked bimetric theory to μ is open. Theorem 4(v) places the ring $\mu = \frac{1}{2}$ at $\sqrt{2GM/a_0}$; the radius $\sqrt{GM/a_0}$ quoted as the processing boundary in Paper I is where $x = \varphi$.

References

[1] S. Nelson, *Gravity Field Dynamics: Field Closure of the Tetrad Frame*, Letter 3 (2026).

[2] S. Nelson, *Covariant completion*, supplementary note to Letter 3 (2026).

[3] S. Nelson, *A Covariant Theory Without Dark Matter*, Paper II (2026), doi:10.5281/zenodo.18360354.

[4] S. Nelson, *The energy functional of the constitutive law* (2026), `proofs/aqual-energy-functional/`.

[5] S. Nelson, *The Other Half of Gravity: Torsion and the Two-Chamber Structure of the Tetrad Frame*, Letter 2 (2026).

[6] S. Nelson, *The constitutive law from exchange symmetry* (2026), `proofs/constitutive-law-exchange-symmetry/`.

[7] S. Nelson, *The Radial Acceleration Relation from a Zero-Parameter Covariant Action* (2026).

[8] S. Nelson, *Gravity Field Dynamics — Coles Notes* (2026), §§3, 5.4, 12.

[9] S. Nelson, *Einstein's equation as the TetA partition* (2026).

[10] S. Nelson, *Documentation: The Field Equation; Covariant Completion* (2026), `docs/02-gravitation/`.

[11] S. Nelson, *One Equation, Two Scales: Tully–Fisher Evolution and the Hubble Constant*, Paper X (2026).

[12] S. Nelson, *Decoding Mass from Observed Rotation Curves* (2026), doi:10.5281/zenodo.18721752.

[13] S. Nelson, *The Gravitational Constitutive Law*, Paper I (2026), doi:10.5281/zenodo.18332595.

[14] J. Bekenstein and M. Milgrom, "Does the missing mass problem signal the breakdown of Newtonian gravity?", *Astrophys. J.* **286** (1984) 7.

[15] M. Milgrom, "A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis", *Astrophys. J.* **270** (1983) 365.